

Manaswi Saha

HCI RESEARCH SCIENTIST

Accenture Labs, 415 Mission St., San Francisco, CA - 94105, USA

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Overview

An Applied R&D Scientist with expertise in human-centered technology design and development and multi-disciplinary research and 12+ years of experience, leveraging emerging technologies for building innovative AI-driven immersive systems and human experiences.

Core Expertise: future of work, product and service innovation, business transformation, accessibility, healthcare, sustainability

- Identify, prioritize, and lead research agendas and roadmaps that generate insights to inform technology innovation strategies for business reinvention and product innovation
- Identify and utilize best practices in human-centered technology design and development for business use cases by conducting user studies and building proof-of-concept prototypes

Technical Areas of Interest: Human-AI/agent interaction and experiences • AI assistants — multimodal conversational AI, always-on contextual AI, emotion AI • AR — visual, audio, and geospatial

Specialties: human-computer interaction, user-centered design, design space development, computer-mediated communication, emerging tech, domain-specific problem discovery and understanding, user studies, interviews, storytelling, cross-functional teams/collaborations and problems

Education

University of Washington

Seattle, USA

PH.D. IN COMPUTER SCIENCE AND ENGINEERING • M.S. IN COMPUTER SCIENCE AND ENGINEERING

September 2017 - August 2022

Advisor: Prof. Jon Froehlich • Google PhD Fellowship 2020 - 2022 • Dissertation: *Designing Interactive Data-Driven Tools for Understanding Urban Accessibility at Scale*

University of Maryland

College Park, USA

PH.D. IN COMPUTER SCIENCE - transferred to University of Washington

August 2015 - August 2017

Advisor: Prof. Jon Froehlich • Dean's Fellowship 2015 - 2017

Vellore Institute of Technology (VIT) University

Vellore, India

MASTER OF COMPUTER APPLICATIONS (MCA)

July 2009 - May 2012

Top 3 in the MCA batch of 2012

University of Mumbai

Mumbai, India

BACHELOR OF SCIENCE - INFORMATION TECHNOLOGY (B.Sc.IT)

June 2006 - April 2009

Top 3 in the B.Sc.IT batch of 2009

Research and Industry Experience

Accenture Labs

San Francisco, USA

ASSOCIATE PRINCIPAL RESEARCHER • DIGITAL EXPERIENCES RESEARCH GROUP

September 2022 - present

Team Lead (s): Alex Kass, Mirjana Spasojevic, and Mike Kuniavsky

Using Human-AI Interaction Lens to Apply Emerging Technologies for Business Problems

Lead HCI Researcher across projects utilizing HAI lens to study and inform enterprise applications • Creating new AI-assisted workflows and pipelines for diverse worker profiles and industry verticals (e.g., utilities, healthcare, media and entertainment) • Exploring multimodal conversational AI agents, biosensing, and applied neuroscience for various business use cases (e.g., workforce training, coding, filmmaking, document understanding, policy compliance) • Conducting user studies and developing proof-of-concept prototypes to demonstrate different types of AI assistants • Through client presentations, translating scientific and research findings to business contexts, demonstrate value, and inform decisions in the downstream business innovation and client delivery units • Working with \$1B business clients on technology innovation projects

Designing Person-aware AI Tools to Support Knowledge Work in Cross-Disciplinary Contexts

Built and leading the research agenda to create generative AI-assisted tools for supporting cross-disciplinary knowledge exchange • Building and testing prototypes using LLMs and web-based tools • Communicating with industry leaders on the value of these tools across a variety of business use cases • Published several academic research papers as the Principal Investigator • Mentored 5 PhD students during summer internships

• **Paper(s):** P.12, P.13

Designing Audio-AR based Guidance Tools to Support Physical Tasks

Built and leading the research agenda to create tools for supporting physical task guidance using Audio AR technology • Leading an ongoing collaboration with Cornell Tech on exploring the uses of audio-only AR interactions for healthcare tasks • Research advisory role for 3 PhD student collaborators and in-lab summer interns • Communicating the business value via client presentations and external blogs • **Paper(s):** EA.11, B.1-B.3

Exploring Bystander Human Reactions for Robot Error Detection

Collaboration with Cornell Tech on HRI research for using human responses to detect robot errors • Advised PhD students on academic publications

• **Paper(s):** P.10-11, EA.9-10

University of Washington

Seattle, USA

GRADUATE RESEARCH ASSISTANT • MAKEABILITY LAB

September 2017 – July 2022

Advisor: Prof. Jon Froehlich • Collaborator(s): Prof. Jeffrey Heer (UW)

Designing Interactive Tools for Visualizing and Modeling Urban Accessibility At Scale

Led formative work around interactive visualization tools for urban accessibility using Project Sidewalk data • Built geovis prototypes using mapbox, leaflet, d3, and kepler.gl • Designed and ran qualitative studies (N=25) to understand multi-stakeholder needs and sensemaking practices of stakeholders, including policymakers, disability advocates, government officials, and people with disabilities • **Paper(s)**: P.9, EA.4-5, EA.7-8, T.1

Urban Accessibility as a Socio-Political Problem

Led a formative interview study with five stakeholders (N=25)—policymakers, transportation department officials, people with disabilities, caregivers, and accessibility advocates—to understand the socio-political challenges impeding accessible infrastructure development and the role of technology to facilitate cross-stakeholder interactions • **Paper(s)**: P.8, T.1

Project Sidewalk: Enabling Crowd-powered Sidewalk Accessibility Data Collection

Lead researcher and engineer of Project Sidewalk (PS), a Google Street View based crowdsourcing tool • Led a 10-person R&D team to develop, publicly deploy, and maintain PS tool • Wrote 15.5K lines of code for HTML/CSS/JS frontend and Java/Scala/PostgreSQL backend • Generated dataset with 250,000+ labels over a 18-month deployment in Washington DC • Designed and ran interview studies with multiple stakeholders (N=14) in the government and the disability community • **Paper(s)**: P.6, EA.3, EA.5-6, T.1 • <http://projectsidewalk.io>

Autodesk Research

Toronto, Canada

RESEARCH INTERN • HCI/VIS: USER INTERFACE RESEARCH GROUP • REMOTE INTERNSHIP

June 2020 – October 2020

Mentor: Justin Matejka

Understanding people's perception of metrics, especially environmental sustainability metrics

Explored how people assessed different metrics to gain insights towards effective communication of sustainability information • Built a HTML/CSS/JS web app for an online study with 50+ sustainability experts and novices on people's abilities in making assessments for length, weight, cost, power, and carbon footprint

Microsoft Research

Redmond, USA

RESEARCH INTERN • ABILITY AND ENABLE GROUPS

June 2018 – September 2018

Mentors: Meredith Ringel Morris, Ed Cutrell, Alex Fiannaca, and Melanie Kneisel

Last-few-meters Wayfinding Challenge for People with Visual Impairments

Engineered Landmark AI, a mobile app prototype using navigational 3D-audio and computer vision algorithms for addressing the last-few-meters challenge in GPS systems • Conducted a design-probe study to understand and address wayfinding challenges for visually impaired users • Ran a survey and an interview study with 12 participants to create the design space for AI-driven navigation tools • **Paper(s)**: P.7

Adobe Research

San Jose, USA

RESEARCH INTERN • BIG DATA EXPERIENCE LAB

May 2016 – August 2016

Mentors: Tom Jacobs and David Tompkins

Beacon-based Personalized Information Delivery

Created an ecosystem design and prototype for Bluetooth beacon-based personalized information delivery for the digital marketing domain that bridges the online world with the physical (brick and mortar stores) • **Patent** filed as the lead in Feb 2017 (PA.1)

University of Maryland

College Park, USA

GRADUATE RESEARCH ASSISTANT • MAKEABILITY LAB

August 2015 – August 2017

Advisor: Prof. Jon Froehlich

Novice Thermography

Explored the use of thermal cameras mounted on smartphones by novices (e.g., DIY enthusiasts) to conduct thermography in homes • Analyzed the interview data for a 4-week field study with 10 participants • **Paper(s)**: P.5, EA.2

Indraprastha Institute of Information Technology Delhi

New Delhi, India

RESEARCH ASSOCIATE • MOBILE AND UBIQUITOUS COMPUTING LAB

November 2012 – July 2015

Advisor: Prof. Amarjeet Singh • Collaborator(s): Prof. Anind Dey (UW), Prof. Yuvraj Agarwal (CMU), Prof. Pushpendra Singh (IIIT-Delhi)

Personal Energy Monitoring in Smart Homes

Led research that explored using smartphone sensors with smart electricity meter for inferring and aggregating daily energy consuming activities to individuals • Engineered EnergyLens+, a real-time energy apportionment and feedback system for smart living spaces to provide real-time personal energy usage information; a Python/Django/MySQL server and an Android app for visualizing feedback • Evaluated the system with a small-scale deployment in single and multi-occupant homes over 2 weeks • **Paper(s)**: P.2, P.3

SensorAct: An Occupant-aware Middleware for Building Energy Management

Led a major engineering effort that allowed SensorAct, a Java/MongoDB-based middleware to operate hardware sensor modules, mounted with ambient environmental sensors, through an online interface and share their access and control with building occupants • **Paper (s)**: P.4, EA.1

Publications

PEER-REVIEWED CONFERENCE PAPERS

- 2025 P.12 **Steering AI-Driven Personalization of Scientific Text for General Audiences**
T. Kim, D. Agarwal, J. Ackerman, **M. Saha**
CSCW: ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing. Bergen, Norway.
- P.11 **Exploring the Design Space of Real-time LLM Knowledge Support Systems: A Case Study of Jargon Explanations**
Y. Liu, A. Shah, J. Ackerman, **M. Saha**
CHI: ACM SIGCHI Conference on Human Factors in Computing Systems. Yokohama, Japan. (Acceptance Rate: 24.9%)
- 2024 P.10 **Bad Idea, Right? Exploring Anticipatory Human Reactions for Outcome Prediction in HRI**
M. T. Parreira, S. G. Lingaraju, A. G. Ramirez-Aristizabal, A. Bremers, **M. Saha**, M. Kuniavsky, W. Ju
ROMAN: IEEE International Conference on Robot and Human Interactive Communication. Los Angeles, USA.
- 2022 P.9 **Visualizing Urban Accessibility: Investigating Multi-stakeholder Perspectives through a Map-based Design Probe Study**
M. Saha, S. Patil, E. Cho, E. Y. Cheng, C. Horng, D. Chauhan, R. Kangas, R. McGovern, A. Li, J. Heer, and J. E. Froehlich
CHI: ACM SIGCHI Conference on Human Factors in Computing Systems. New Orleans, USA. (Acceptance Rate: 24.7%)
- 2020 P.8 **Urban Accessibility as a Socio-Political Problem: A Multi-Stakeholder Analysis**
M. Saha, D. Chauhan, S. Patil, R. Kangas, J. Heer, and J. E. Froehlich
CSCW: ACM Conference on Computer-Supported Cooperative Work and Social Computing. Virtual. (Acceptance Rate [minor rev.]: 7.9%)
- 2019 P.7 **Closing the Gap: Designing for the Last-Few-Meters Wayfinding Problem for People with Visual Impairments**
M. Saha, A. J. Fiannaca, M. Kneisel, E. Cutrell, M. R. Morris
ASSETS: ACM SIGACCESS Conference on Computers and Accessibility. Pittsburgh, USA. (Acceptance Rate: 26%)
- P.6  **Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data At Scale**
M. Saha, M. Saugstad, H. Maddali, A. Zeng, R. Holland, S. Bower, A. Dash, S. Chen, A. Li, K. Hara, J. Froehlich
CHI: ACM SIGCHI Conference on Human Factors in Computing Systems. Glasgow, UK. (Acceptance Rate: 23.8%) • **Best Paper Award (Top 1%)**
- 2017 P.5 **Exploring Novice Approaches to Smartphone-Based Thermographic Energy Auditing: A Field Study**
M. L. Mauriello, **M. Saha**, E. Brown, J. E. Froehlich
CHI: ACM SIGCHI Conference on Human Factors in Computing Systems. Denver, USA. (Acceptance Rate: 25%)
- 2015 P.4 **SensorAct: A Decentralized and Scriptable Middleware for Smart Buildings**
P. Arjunan*, **M. Saha***, H. Choi, M. Gulati, A. Singh, P. Singh, M. B. Srivastava
IEEE UIC: IEEE International Conference on Ubiquitous Intelligence and Computing. Beijing, China. (Acceptance Rate: 30.6%)
- 2014 P.3 **WattShare: Detailed Energy Apportionment in Shared Living Spaces within Commercial Buildings**
S. Thakur, **M. Saha**, A. Singh, Y. Agarwal
BuildSys: ACM International Conference on Embedded Systems for Energy-Efficient Buildings. Memphis, USA. (Acceptance Rate: 27%)
- P.2 **EnergyLens: Combining Smartphones with Electricity Meter for Accurate Activity Detection and User Annotation**
M. Saha, S. Thakur, A. Singh, Y. Agarwal
e-Energy: ACM International Conference on Future Energy Systems. Cambridge, UK. (Acceptance Rate: 20%)
- 2012 P.1 **Bandwidth Management Framework for Multicasting in Wireless Mesh Networks**
M. Saha, P. V. Krishna
IJIEE: International Journal of Information and Electronics Engineering. Vol. 2, No. 3.

DISSERTATION

- 2022 T.1 **Designing Interactive Data-Driven Tools for Understanding Urban Accessibility at Scale**
M. Saha
University of Washington. Computer Science & Engg. PhD Dissertation. Seattle, USA.

PEER-REVIEWED POSTERS, EXTENDED ABSTRACTS, DOCTORAL COLLOQUIUM, AND WORKSHOPS

- 2024 EA.11 **Situated Conversational Agents for Task Guidance: A Preliminary User Study**
A. Bremers*, **M. Saha***, A. G. Ramirez-Aristizabal
ACM CUI: Proceedings of the 6th ACM Conference on Conversational User Interfaces. Luxembourg City, Luxembourg.
- EA.10 **A Study on Domain Generalization for Failure Detection through Human Reactions in HRI**
M. T. Parreira, S. G. Lingaraju, A. G. Ramirez-Aristizabal, **M. Saha**, M. Kuniavsky, Wendy Ju
HRI Workshop: Social Signal Modeling in Human-Robot Interaction. Boulder, USA.
- 2023 EA.9 **Bad Idea? Exploring Anticipatory Human Reactions for Outcome Prediction**
M. T. Parreira, S. G. Lingaraju, A. G. Ramirez-Aristizabal, A. Bremers, **M. Saha**, M. Kuniavsky, W. Ju
NERC: Northeast Robotics Colloquium. New Haven, USA.
- 2022 EA.8 **The Future of Urban Accessibility for People with Disabilities: Data Collection, Analytics, Policy, and Tools**
J. E. Froehlich, Y. Eisenberg, M. Hosseini, F. Miranda, M. Adams, A. Caspi, H. Dieterich, ..., **M. Saha** et al.
ASSETS: ACM SIGACCESS Conference on Computers and Accessibility. Workshop. Athens, Greece.

- 2021 EA.7 **The Future of Global-Scale Spatial Data Collection and Analyses on Urban (in)Accessibility for People with Disabilities**
J. E. Froehlich, F. Miranda, M. Hosseini, N. Bolten, A. Caspi, R. M. Cesar Jr., H. Dieterich, Y. Eisenberg, V. Pineda, **M. Saha** et al.
Spatial Data Science Symposium. Virtual.
- 2020 EA.6 **Towards Mapping and Assessing Sidewalk Accessibility Across Sociocultural and Geographic Contexts**
J. E. Froehlich, M. Saugstad, **M. Saha**, M. Johnson
AVI Workshop: Data4Good - Designing for Diversity and Development. Ischia, Italy.
- 2019 EA.5 **Interactive Computational Tools for Assessing and Understanding Urban Accessibility At Scale**
M. Saha
ASSETS: ACM SIGACCESS Conference on Computers and Accessibility. Doctoral Colloquium. SIGACCESS Newsletter. Pittsburgh, USA.
- 2018 EA.4 **Interactively Modeling and Visualizing Neighborhood Accessibility at Scale: An Initial Study of Washington DC**
A. Li, **M. Saha**, A. Gupta, J. E. Froehlich
ASSETS: ACM SIGACCESS Conference on Computers and Accessibility. Poster & Demo. Galway, Ireland.
- 2017 EA.3 **A Pilot Deployment of an Online Tool for Large-Scale Virtual Auditing of Urban Accessibility**
M. Saha, K. Hara, S. Behnezhad, A. Li, M. Saugstad, H. Maddali, S. Chen, J. E. Froehlich
ASSETS: ACM SIGACCESS Conference on Computers and Accessibility. Poster & Demo. Baltimore, USA.
- 2016 EA.2 **The Future Role of Thermography in Human-Building Interaction**
M. L. Mauriello, M. Dahlhausen, E. Brown, **M. Saha**, J. E. Froehlich
CHI Workshop: Future of Human-Building Interaction. San Jose, USA.
- 2013 EA.1 **SensorAct: Design and Implementation of Fine-grained Sensing and Control Sharing in Buildings**
P. Arjunan, **M. Saha**, M. Gulati, N. Batra, A. Singh, P. Singh
NSDI: USENIX Symposium on Networked Systems Design and Implementation. Poster. Lombard, USA.

BLOGS

- 2025 B.4 **What We Say, Think, and Feel: AI & Biosensing's Transformative Capabilities for Creators**
A. G. Ramirez-Aristizabal, **M. Saha**, M. Spasojevic, A. Naghi, K. Glass
Labs Notebook Medium Blog Series. Blog. San Francisco, USA.
- 2023 B.3 **Audio AR — Part 3: Acoustic Digital Twin**
W. Ju, **M. Saha**, M. Kuniavsky, D. Goedicke
Labs Notebook Medium Blog Series. Blog. San Francisco, USA.
- B.2 **Audio AR — Part 2: Acoustic Sensing**
D. Goedicke, **M. Saha**, W. Ju, M. Kuniavsky
Labs Notebook Medium Blog Series. Blog. San Francisco, USA.
- B.1 **Audio AR: An Introduction – Part 1: Towards Hands-Free Eyes-Free Interaction**
M. Saha, W. Ju, M. Kuniavsky, D. Goedicke
Labs Notebook Medium Blog Series. Blog. San Francisco, USA.

Patents

- 2017 PA.1 **Digital Content Output Control in a Physical Environment Based on a User Profile**
Manaswi Saha, Tom Jacobs, David Tompkins, Peter Fransen
Adobe Research. Filed in February 2017. Patent Pending.

Skills

Programming	Python, Java, C, C++, Shell Scripting
AI Tools	Cursor, v0 / Vercel, Gemini, Open AI tools
Front-end	HTML/CSS, JavaScript, React
Web Frameworks	Play, Django
Backend Databases	PostgreSQL, MySQL, MongoDB
Visualization Tools	mapbox, d3, Tableau, kepler.gl
Data Analysis and Applied ML	Python: pandas, numpy, matplotlib, sklearn
Crowdsourcing Data	Mechanical Turk, Prolific
Other Tools	Latex, Github, Figma, IntelliJ IDEA, Eclipse
User Research	Interviews, surveys, thematic analysis, affinity diagramming, stakeholder analysis

Intern Mentorship

Accenture Labs Yuhan Liu (Princeton, PhD), Taewook Kim (Northwestern, PhD), Dhruv Agarwal (Cornell, PhD), Alexandra Bremers (Cornell Tech, PhD), Jiachen Li (Northeastern, PhD), Aadit Shah (Ohio State, Undergrad Senior), Anna Nguyen (Rice, Undergrad Junior)

University of Washington Evie Yu-Yen Cheng (Masters), Emily Cho (Masters), Chris Horng (Masters), Devanshi Chauhan (Masters), Rachel Kangas (Masters), Siddhant Patil (Masters), Richard McGovern (Masters), Aileen Zeng (Sophomore), Johnson Kuang (Freshman)

University of Maryland Anthony Li (Senior), Aditya Dash (Senior), Steven Bower (Senior), Maria Furman (Senior), Chirag Shankar (Junior), Sage Chen (Sophomore), Ji Hyuk Bae (Freshman), Ryan Holland (High School Senior)

IIIT-Delhi Shailja Thakur (Masters), Vedant Das Swain (Senior)

Honors & Awards

2020 Google PhD Fellowship in HCI	To support thesis work on interactive tools for urban accessibility	Seattle, USA
2019 Amazon Catalyst Award	To support thesis work on interactive tools for urban accessibility	Seattle, USA
Best Paper Award	ACM CHI 2019 for Project Sidewalk	Glasgow, UK
ACM-W Scholarship	ACM CHI 2016	San Jose, USA
CRA-W Grad Cohort Participation Grant	Grad Cohort Workshop 2016	San Diego, USA
Dean's Fellowship	University of Maryland, College Park (2015 – 2017)	College Park, USA
Microsoft Research India Travel Grant	To present at ACM e-Energy 2014	Cambridge, UK
Merit Scholarship	VIT University (2009 – 2012) – awarded all three years – top 3 (of 120) students	Vellore, India
Certificate of Merit	B.Sc.IT batch (2006 – 2009) – awarded all three years – top 3 (of 60) students	Mumbai, India
Certificate of Merit	Best Bachelor's (B.Sc.IT) Project	Mumbai, India

Talks

- 2025 T.18 **Transforming Human Experiences: Designing for Real-World Impact using Data x HCI x AI**
Guest lecture for CPSC 444 Advanced Methods For HCI: Industry Panel, University of British Columbia. Virtual. April 2025 • **Invited Talk**
- 2022 T.17 **Visualizing Urban Accessibility: Investigating Multi-stakeholder Perspectives through a Map-based Design Probe Study**
ACM Conference on Human Factors in Computing Systems (CHI). New Orleans, USA. May 2022
- 2021 T.16 **Visualizing Urban Accessibility: Understanding Sensemaking Processes across Multiple Stakeholders**
IEEE VIS 2021 Workshop on Visualization for Social Good. Virtual. October 2021
- T.15 **Designing Interactive Computational Tools for Understanding Urban Accessibility: Exploring the Nexus of Urban Informatics x Accessibility**
Guest lecture for Computing for Social Good, University of Delaware. Virtual. April 2021 • **Invited Talk**
- T.14 **Designing Interactive Tools for Urban Accessibility: A Socio-Political and Socio-Technical Perspective**
Guest Lecture for Population, Economy, and Society, IIT Kanpur. Virtual. April 2021 • **Invited Talk**
- 2020 T.13 **Designing Interactive Tools for Understanding Urban Accessibility**
NWFSC Monster Seminar Jam Series at Northwest Fisheries Science Center, NOAA. Virtual. October 2020 • **Invited Talk**
- T.12 **Urban Accessibility as a Socio-Political Problem: A Multi-Stakeholder Analysis**
ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW). Virtual. October 2020
- T.11 **Urban Accessibility as a Socio-Political Problem: A Multi-Stakeholder Analysis**
Accessibility Colloquium, University of Washington. Seattle, USA. October 2020
- T.10 **Project Sidewalk: Mapping the Accessibility of the Physical World At Scale Using Interactive Computational Tools**
Google Tech Talk. Seattle, USA. January 2020 • **Invited Talk**
- 2019 T.9 **Closing the Gap: Designing for the Last-Few-Meters Wayfinding Problem for People with Visual Impairments**
ACM International SIGACCESS Conference on Computers and Accessibility (ASSETS). Pittsburgh, USA. October 2019
- T.8 **Project Sidewalk: Mapping the Accessibility of the Physical World At Scale Using Interactive Computational Tools**
Washington State Ridesharing Organization (WSRO) Annual Conference. Spokane, USA. September 2019 • **Keynote Speaker**
- T.7 **Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data At Scale**
ACM Conference on Human Factors in Computing Systems (CHI). Glasgow, UK. May 2019 • **Best Paper Award (Top 1%)**
- 2017 T.6 **Bridging the Pedestrian Accessibility Informational Gap: User-Facing Applications and Large-Scale Virtual Auditing**
Affiliates Research Day, University of Washington. Seattle, Washington. November 2017
- T.5 **Project Sidewalk: Assessing Urban Accessibility using Crowdsourcing and Google Street View**
HCIL Symposium, University of Maryland. College Park, USA. May 2017
- T.4 **Project Sidewalk: Characterizing Physical World Accessibility at Scale**
WalkHackNight II. Arlington, USA. February 2017 • **Invited Talk**
- 2016 T.3 **Interactive Computational Tools For Accessibility**
Diversity in Computing Summit, University of Maryland. College Park, USA. November 2016
- T.2 **Tech+Design: Interaction Design For A Purpose**
Technica: Tech+X Talk Series. College Park, USA. November 2016
- 2014 T.1 **EnergyLens: Combining Smartphones with Electricity Meter for Accurate Activity Detection and User Annotation**
Fifth International Conference on Future Energy Systems (ACM e-Energy). Cambridge, UK. June 2014

Press and Media Coverage

Apr 2022	Accessible Sidewalks for Inclusive Cities	World Bank's Sustainable Cities Blog	↗
Dec 2021	Behind the effort to make sidewalks accessible	NPR's "Here and Now"	↗
May 2021	Make The World Better With One Of These Nine Ideas	New York Times	↗
Oct 2020	Manaswi Saha wins 2020 Google Fellowship for advancing computing research with social impact	Allen School News	↗
Oct 2020	Manaswi Saha, Lead Graduate Student for Project Sidewalk, Wins Google Fellowship	Urbanalytics News	↗
Oct 2020	Announcing the 2020 Google PhD Fellows	Google AI News	↗
Dec 2019	Can Mapping Sidewalks Increase Ridership?	National Center for Mobility Management	↗
Oct 2019	Manaswi Saha wins Amazon Catalyst Award to develop techniques for visualizing urban accessibility at scale	Allen School News	↗
Oct 2019	Amazon Catalyst announces new round of Fellows	CoMotion News	↗
May 2019	Graduate students want to map the world's sidewalks for their accessibility	KIRO 7 News	↗
May 2019	Players label accessibility of city streets in new game	IEEE GlobalSpec Electronics 360	↗
Apr 2019	Newberg streets second in nation to be studied with new accessibility metric	The Newberg Graphic	↗
Apr 2019	Seattle's got terrible sidewalks. You can help fix them.	Crosscut	↗
Apr 2019	Project Sidewalk helps users map accessibility around Seattle, other cities	UW News	↗
Feb 2017	Make D.C.'s sidewalks more accessible with this crowd-sourced map	Curbed DC	↗
Feb 2017	University of Maryland project looks to crowdsource an accessibility map of DC's sidewalks	Mobility Lab	↗
Jan 2017	Clear the Way - UMD Computer Scientists Seek Public Help Mapping Sidewalk Accessibility	Terp Magazine	↗
Nov 2016	How Project Sidewalk is making DC more accessible	Technical.ly	↗
Nov 2016	New tool makes DC sidewalks more accessible for everyone	WUSA9 News	↗
Oct 2016	A UMD team made an app highlighting D.C. areas inaccessible to people with disabilities	The Diamondback	↗
Oct 2016	You Can Help Map the Accessibility of the World	Next City	↗
Sep 2016	Missing sidewalks? There's an app for that	Greater Greater Washington	↗

Teaching Experience

University of Washington

Seattle, USA

GRADUATE TEACHING ASSISTANT

January 2018 – June 2020

Paul G. Allen School of Computer Science and Engineering

CSE482A: Capstone Software Design to Empower Underserved Populations

Spring 2020

CSE599H: Crowdsourcing, Citizen Science, and Large-scale Online Experimentation

Winter 2020

CSE599S: The Future of Access Technologies

Fall 2019

CSE441: Advanced HCI: Advanced User Interface Design, Prototyping, And Evaluation

Spring 2018, Spring 2019

CSE440: Introduction to HCI

Winter 2018, Fall 2018, Winter 2019

University of Maryland

College Park, USA

GRADUATE TEACHING ASSISTANT

August 2015 – May 2016

Department of Computer Science

CMSC132: Object-Oriented Programming II

Spring 2016

CMSC131: Object-Oriented Programming I

Fall 2015

Service

2025	ACM CHI 2025 Reviewer
	ACM ICMI 2025 Reviewer
2024	ACM CHI 2024 Reviewer
	ACM ICMI 2024 Reviewer
2023	University of California Berkeley CS Scholars - Panelist
	ACM CHI 2024 Reviewer
	ACM ASSETS 2023 Reviewer
2022	University of California Berkeley CS Scholars - Panelist
	ASSETS 2022 Workshop Organizing Team
	CHI 2023 Reviewer
2021	SDSS 2021 Workshop Organizing Team
	ACM CHI 2022 Reviewer
	IEEE VIS 2022 Reviewer
2020	ACM CHI 2021 Reviewer
	ACM CSCW 2020 Reviewer - Special Recognition for Outstanding Review
2019	ACM CHI 2020 Reviewer
	Incoming UW CSE PhD Students' Mentor
2018	DUB PhD Student Retreat Co-organizer
	UW HCI Skillshares Organizer
	ACM CHI 2018 Student Volunteer
2016	Adobe Girls Who Code Mentor
	ACM CHI 2016 Student Volunteer